

Absorbing Boundary Conditions for an Elongated Body

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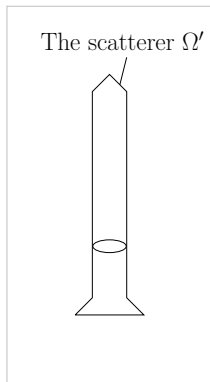


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Acoustic scattering of an elongated obstacle Ω'



The considered domain Ω

$$\Omega = \mathbb{R}^3 \setminus \Omega'$$

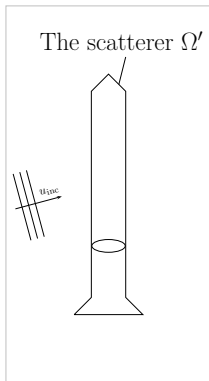
We aim in computing the phasor of the **total pressure field** (**harmonic regime**)

$$p_{\text{tot}}(\mathbf{x}, t) = \Re(u_{\text{tot}}(\mathbf{x}) \exp(-i\omega t))$$

generated by an **incident plane wave**

$$u_{\text{inc}}(\mathbf{x}, t) = \exp(i\mathbf{k} \cdot \mathbf{x})$$

Acoustic scattering of an elongated obstacle Ω'



The acoustic system writes

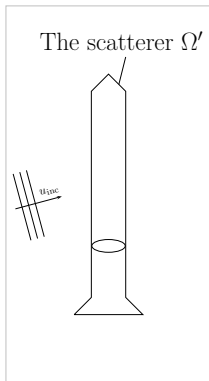
$$\left\{ \begin{array}{l} u_{\text{tot}} \in H_{\text{loc}}^1(\Omega), \\ \Delta u_{\text{tot}}(\mathbf{x}) + k^2 u_{\text{tot}}(\mathbf{x}) = 0 \text{ in } \Omega, \\ \frac{\partial u_{\text{tot}}}{\partial n}(\mathbf{x}) = 0 \text{ on } \partial\Omega, \\ u_{\text{tot}}(\mathbf{x}) - u_{\text{inc}}(\mathbf{x}) \text{ is outgoing.} \end{array} \right.$$

outgoing $\iff$ absorption principle

Finite element approximation is not adapted for **infinite domains**.

We need to bound computational domain

Acoustic scattering of an elongated obstacle Ω'



The acoustic system writes

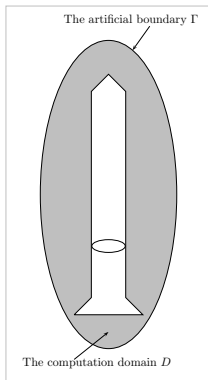
$$\left\{ \begin{array}{l} u \in H_{loc}^1(\Omega), \\ \Delta u(\mathbf{x}) + k^2 u(\mathbf{x}) = 0 \text{ in } \Omega, \\ \frac{\partial u}{\partial n}(\mathbf{x}) = -\frac{\partial u_{inc}}{\partial n}(\mathbf{x}) \text{ on } \partial\Omega, \\ u(\mathbf{x}) \text{ is outgoing.} \end{array} \right.$$

outgoing $\iff$ absorption principle

Finite element approximation is not adapted for **infinite domains**.

We need to bound computational domain

Acoustic scattering of an elongated obstacle Ω'



The restriction of the scattered field solves

$$\begin{cases} u \in H^1(D), \\ \Delta u(\mathbf{x}) + k^2 u(\mathbf{x}) = 0 \text{ in } D, \\ \frac{\partial u}{\partial n}(\mathbf{x}) = -\frac{\partial u_{\text{inc}}}{\partial n}(\mathbf{x}) \text{ on } \partial\Omega, \end{cases}$$

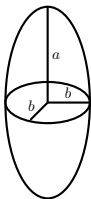
This system should be supplemented with an **A**bsorbing **B**oundary **C**ondition

$$\frac{\partial u}{\partial n}(\mathbf{x}) + DtNu(\mathbf{x}) = 0 \quad \text{on } \Gamma$$

- For large domains, the **computations** are very **costly**
- For small domains, very precise **ABC** are required

- Engquist, Majda (1977): Hyperbolic linear problems for plane surfaces (pseudodifferential calculus)
- Bayliss, Gunzburger, Turkel (1982): ABC (the so-called *BGT conditions*) for the wave equation, spherical shaped boundaries
- Halpern (1991): Parabolic systems
- Szeftel (2004): Pseudodifferential and paradifferential calculus for non linear equations
- Tsynkov (1998), Givoli (2004), Turkel (2008): Reviews

Objective of the paper



Provide very precise **ABC** in the context of prolate ellipsoidal artificial boundaries

$$\Gamma = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \frac{x^2 + y^2}{b^2} + \frac{z^2}{a^2} = 1 \right\}$$

with $b < a$.

Well-suited for elongated scatterers.

- Provide a theoretical background for these conditions
- Test the numerical feasibility and perform numerical tests

- Hanouzet, Sesquès (1992): ABC for the Maxwell's equation on surfaces of arbitrary shapes
- Antoine, Barucq, Bendali (1997): Radiation conditions on surfaces of arbitrary shapes
- Harari, Djellouli (2004): Comparison of BGT and local DtN conditions for circular and spherical shaped boundaries
- Reiner, Djellouli, Harari (2006): BGT conditions for elliptical and ellipsoidal shaped boundaries
- Medvinsky, Turkel, Hetmaniuk (2008): Comparison of different ABC for elliptical and ellipsoidal shaped boundaries
- Barucq, Djellouli, Saint-Guirons (2009): Construction and performance assesment of new local DtN conditions for elongated obstacles

Separation of variables

The derivation of the **ABC** is based on the solution of the Dirichlet exterior problem

$$\left\{ \begin{array}{l} u_{\text{ext}} \in H_{\text{loc}}^1(D'), \\ \Delta u_{\text{ext}}(\mathbf{x}) + k^2 u_{\text{ext}}(\mathbf{x}) = 0 \text{ in } D', \\ u_{\text{ext}}(\mathbf{x}) \text{ given on } \Gamma, \\ u_{\text{ext}}(\mathbf{x}) \text{ is outgoing.} \end{array} \right.$$

with the exterior domain

$$D' = \left\{ \frac{x^2 + y^2}{b^2} + \frac{z^2}{a^2} > 1 \right\} \quad (1)$$

The Dirichlet to Neumann map associates to the value of u_{ext} on Γ (**Dirichlet trace**) the value of $\frac{\partial u_{\text{ext}}}{\partial n}$ (**Neumann trace**).

Separation of variables

$$u_{\text{ext}}(\mathbf{x}) = \sum_{n=0}^{+\infty} \sum_{m=-n}^n u_{m,n} \Psi_{m,n}(\xi) Y_{m,n}(\eta, \varphi) \quad \text{in } D'$$

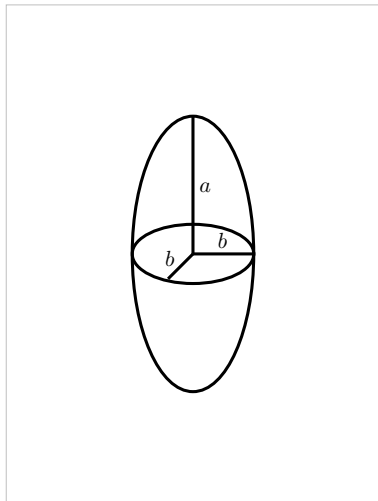
- (ξ, η, φ) the prolate **spheroidal coordinates**
- $\Psi_{m,n}$ the prolate **radial spheroidal functions** of third type
- $Y_{m,n}$ the prolate **angular spheroidal functions**
- the spectral coefficient $u_{m,n}$ are related to the trace of u on Γ

The Dirichlet to Neumann operator is given by

$$\frac{\partial u_{\text{ext}}}{\partial n}(\mathbf{x}) = \frac{b}{f^2} \sum_{n=0}^{+\infty} \sum_{m=-n}^n \frac{Z_{m,n} u_{m,n} Y_{m,n}(\eta(\mathbf{x}), \varphi(\mathbf{x}))}{\sqrt{a^2/f^2 - z^2/a^2}} \quad \text{on } \Gamma.$$

- The $Z_{m,n}$ are the **impedance coefficients**

The prolate spheroidal coordinates



are defined by

$$\begin{cases} x = f \sqrt{(\xi^2 - 1)(1 - \eta^2)} \cos(\varphi) \\ y = f \sqrt{(\xi^2 - 1)(1 - \eta^2)} \sin(\varphi), \\ z = f \xi \eta, \end{cases}$$

with

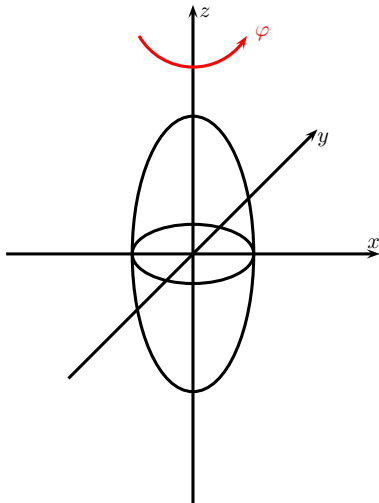
$$\xi \geq 1, -1 \leq \eta \leq 1, -2\pi \leq \varphi \leq 2\pi.$$

φ : angle with respect to the z-axis

ξ : spheroidal radius

η : relative altitude on the spheroid

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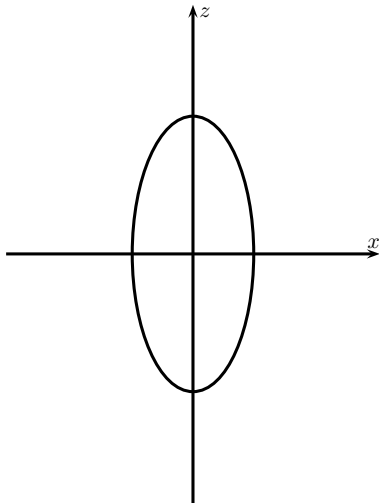
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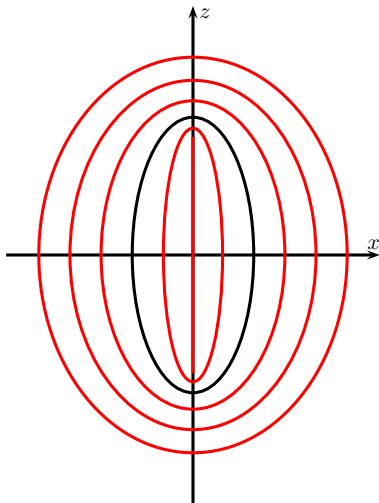
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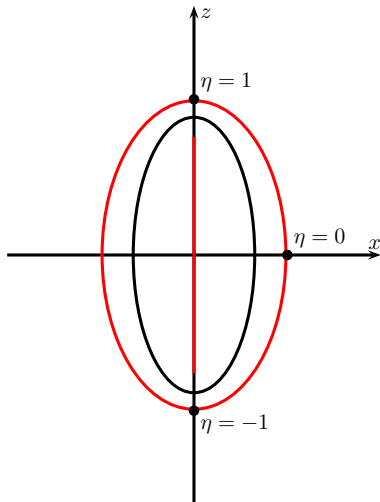
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The radial spheroidal functions

The radial spheroidal functions of third type solve

$$\begin{cases} \frac{\partial}{\partial \xi} \left((\xi^2 - 1) \frac{\partial \psi_{m,n}}{\partial \xi} \right) (c, \xi) + \left(c^2 \xi^2 - \frac{m^2}{\xi^2 - 1} - \lambda_{m,n}(c) \right) \psi_{m,n}(c, \xi) = 0, \\ \frac{d\psi_{m,n}}{d\xi}(c, \xi) - ic\psi_{m,n}(c, \xi) = O_{\xi \rightarrow +\infty} \left(\frac{1}{\xi^2} \right), \end{cases}$$

- defined up to a multiplicative constant
- We fix the multiplicative constant by imposing at infinity

$$\psi_{m,n}(c, \xi) = \frac{\exp(ic\xi)}{\xi} \left(1 + o_{\xi \rightarrow +\infty}(1) \right).$$

The angular spheroidal functions

are the eigenfunctions of a perturbed **Laplace Beltrami** operator

$$\left\{ \begin{array}{l} \text{Find } Y \in H^1(\Gamma) \text{ and } \lambda \in \mathbb{R} \text{ such that for all } v \in H^1(\Gamma) \\ (Y, v)_{\Gamma,1} = \lambda (Y, v)_{\Gamma,0} \end{array} \right. \quad (2)$$

with the bilinear forms

$$\left\{ \begin{array}{l} (u, v)_{\Gamma,0} = \int_{-1}^1 \int_0^{2\pi} u(\eta, \varphi) \bar{v}(\eta, \varphi) d\varphi d\eta \\ (u, v)_{\Gamma,1} = \int_{-1}^1 \int_0^{2\pi} \partial_\eta u(\eta, \varphi) \partial_\eta \bar{v}(\eta, \varphi) (1 - \eta^2) \\ \quad + \partial_\varphi u(\eta, \varphi) \partial_\varphi \bar{v}(\eta, \varphi) \\ \quad + c^2 (\eta^2 + 1) u(\eta, \varphi) \bar{v}(\eta, \varphi) d\varphi d\eta. \end{array} \right. \quad (3)$$

with $c = kf > 0$.

The angular spheroidal functions

Parametrization of the eigenfunctions by

$$n \in \mathbb{N} \quad \text{and} \quad -m \leq n \leq m.$$

- m is related to the number of oscillation in φ
- n is related to the number of oscillation in η

$$Y_{m,n}(\eta, \varphi) = \tilde{Y}_{m,n}(\eta) \exp(im\varphi)$$

with $\tilde{Y}_{m,n}(\eta)$ which solves a 1D eigenvalue problem

$$\int_{-1}^1 \partial_{\eta} \tilde{Y}_{m,n}(\eta) \partial_{\eta} \bar{v}(\eta) (1 - \eta^2) d\eta + \int_{-1}^1 \left(c^2 \eta^2 + c^2 - \lambda_{m,n} \right) \tilde{Y}_{m,n}(\eta) \bar{v}(\eta) d\eta = 0.$$

The angular spheroidal functions

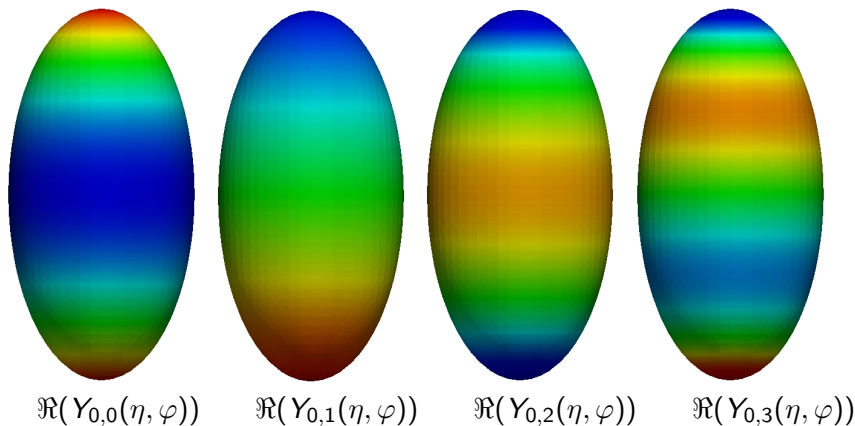


Figure: Real part of $Y_{m,n}$ for $a = 1$, $b = .5$, $k = 1$

The angular spheroidal functions

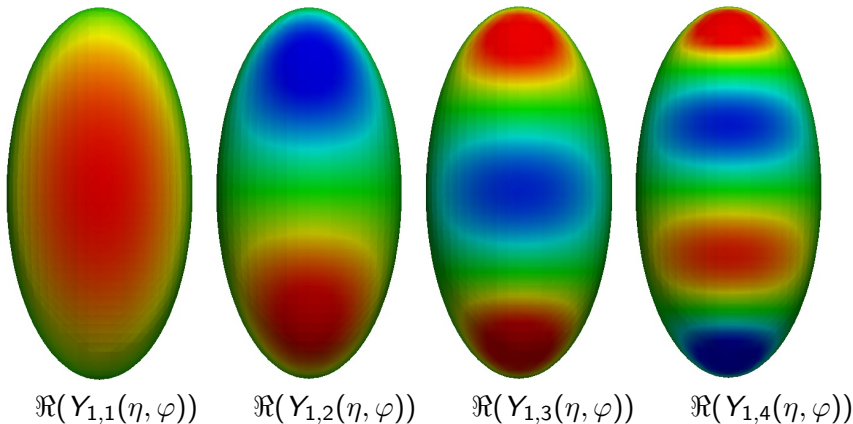


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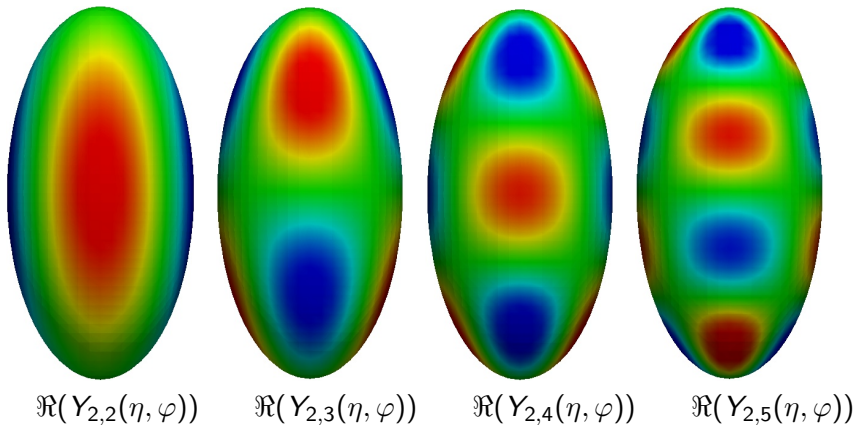


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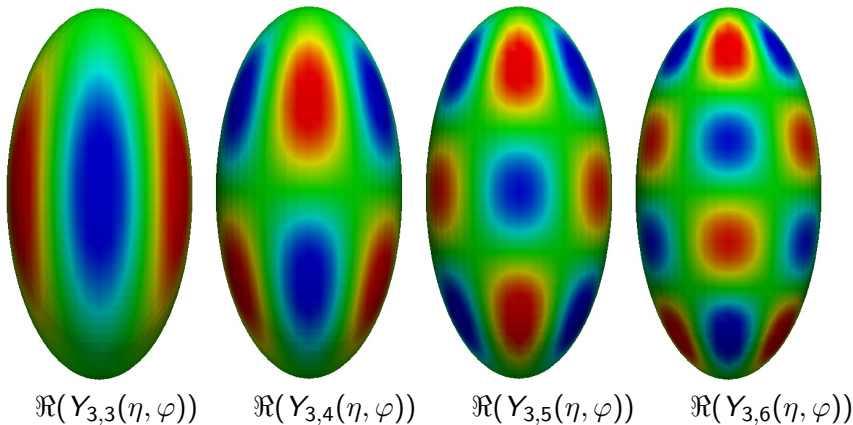


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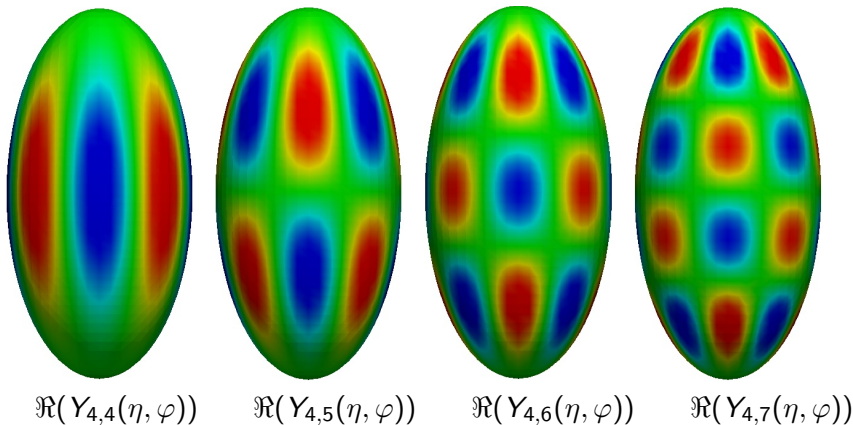


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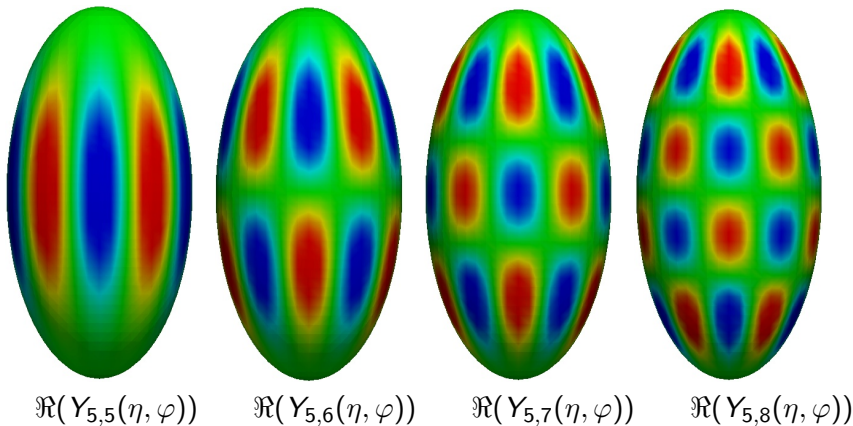


Figure: Real part of $Y_{m,n}$ for $a = 1$, $b = .5$, $k = 1$

The impedance coefficients

They are defined by

$$Z_{m,n}(c, \xi) = \frac{\frac{d\psi_{m,n}}{d\xi}(c, \xi)}{\psi_{m,n}(c, \xi)}$$

and solve a **generalized Riccati equation**

$$\begin{cases} \partial_{\xi} Z_{m,n}(c, \xi) + F_{m,n}(Z_{m,n}(c, \xi), \xi) = 0 & \forall \xi > 1 \\ \lim_{\xi \rightarrow +\infty} Z_{m,n}(c, \xi) = ic \end{cases}$$

with the function $F_{m,n} : \mathbb{C} \times]1, +\infty[\rightarrow \mathbb{C}$ given by

$$F_{m,n}(Z, \xi) = Z^2 + \frac{2\xi}{\xi^2 - 1}Z + \left(c^2 + \frac{c^2 - \lambda_{m,n}(c)}{\xi^2 - 1} - \frac{m^2}{(\xi^2 - 1)^2} \right).$$

The impedance coefficients

They are computed by a **backward Runge-Kutta 4 method** and by imposing a radiation condition

The radiation condition is obtained by expanding $Z_{m,n}$ analytically in the neighborhood of infinity

$$Z_{m,n}(c, \xi) = Z_{m,n}^{asym,p}(c, \xi) + \underset{\xi \rightarrow +\infty}{O}\left(\frac{1}{\xi^{p+1}}\right).$$

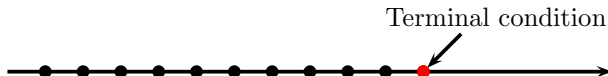
with

$$\left\{ \begin{array}{lcl} Z_{m,n}^{asym,0}(c, \xi) & = & ic \\ Z_{m,n}^{asym,1}(c, \xi) & = & ic - \frac{1}{\xi} \\ Z_{m,n}^{asym,2}(c, \xi) & = & ic - \frac{1}{\xi} + \frac{\lambda_{m,n} - c^2}{2ic} \frac{1}{\xi^2} \\ Z_{m,n}^{asym,3}(c, \xi) & = & ic - \frac{1}{\xi} + \frac{\lambda_{m,n} - c^2}{2ic} \frac{1}{\xi^2} - \frac{\lambda_{m,n} + c^2}{2c^2} \frac{1}{\xi^3} \end{array} \right.$$

Computation of the impedance coefficients

The approximation of $Z_{m,n}$ of order p is defined by

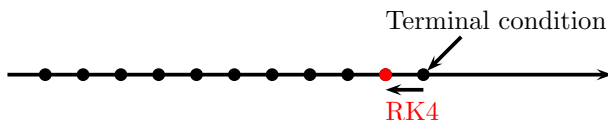
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Computation of the impedance coefficients

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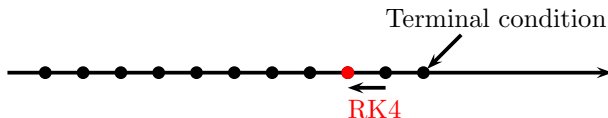
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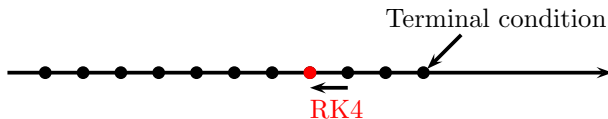
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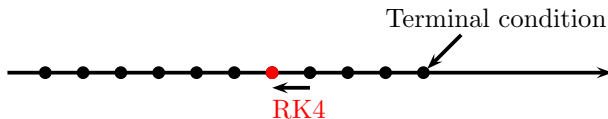
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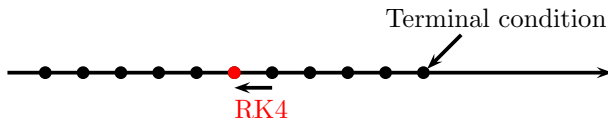
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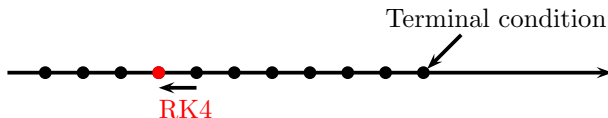
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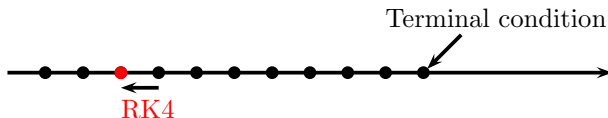
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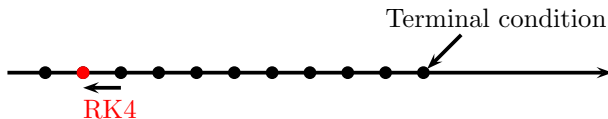
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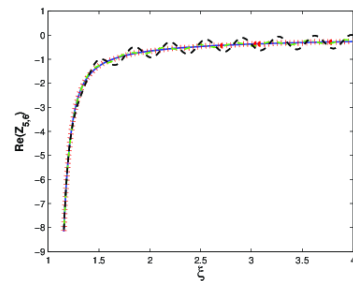
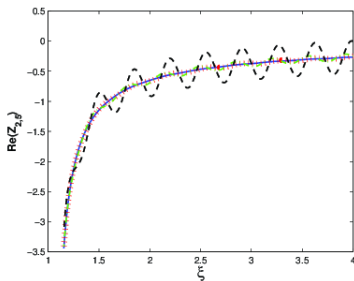
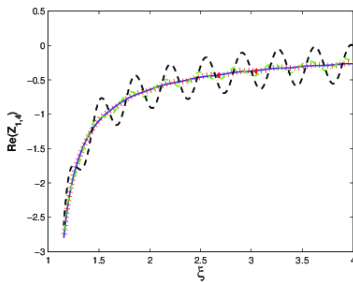
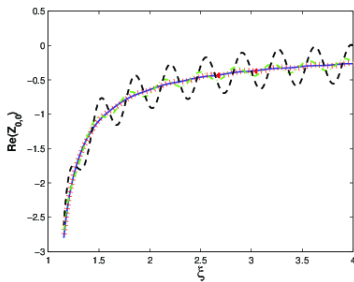
Computation of the impedance coefficients

The approximation of $Z_{m,n}$ of order p is defined by

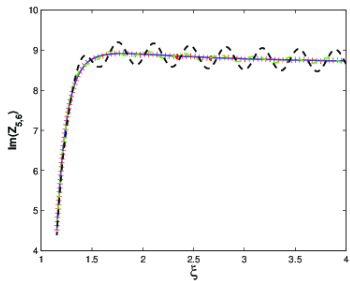
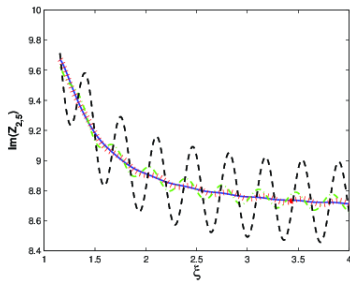
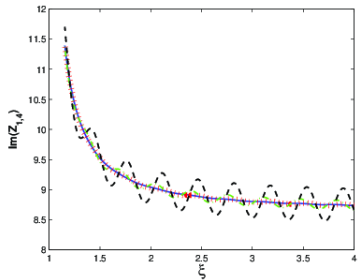
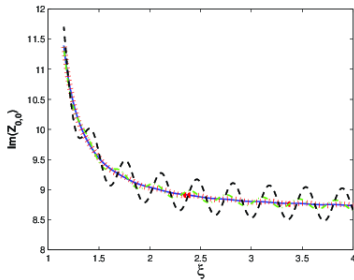
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Computation of the impedance coefficients



Computation of the impedance coefficients



Main properties of the impedance coefficients

Lemma

For all (m,n) , we have

$$\Im Z_{m,n}(c, \xi) > 0.$$

Conjecture

For all (m,n) , we have

$$\Re Z_{m,n}(c, \xi) < 0.$$

A well-posed variational formulation

The solution $u \in H^1(D)$ is the unique function satisfying for all v

$$\int_D \nabla u \cdot \overline{\nabla v} - k^2 u \bar{v} - \frac{b}{f^2} \sum_{\substack{n \in [0, +\infty] \\ -n \leq m \leq n}} Z_{m,n} u_{m,n} \overline{v_{m,n}} = - \int_{\partial\Omega} \frac{\partial u_{\text{inc}}}{\partial n} \bar{v}.$$

with $u_{m,n}$

$$u_{m,n} = \frac{1}{ab} \int_{\Gamma} \frac{u(\mathbf{x}) \overline{Y_{m,n}(\mathbf{x})} ds_{\mathbf{x}}}{\sqrt{a^2/f^2 - z^2/a^2}}.$$

Existence and uniqueness follows from the Fredholm alternative, the unique continuation theorem and from

$$\Im(Z_{m,n}) > 0.$$

- To be implemented this series needs to be **truncated**.
- For some geometry, this formulation is **ill-posed** or **ill-conditioned**

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Our formulation is based on the identity

$$\begin{aligned} b(u, v) &= \sum_{\substack{n \in [0, +\infty] \\ -n \leq m \leq n}} Z_{m,n} u_{m,n} \overline{v_{m,n}} \\ &= \sum_{\substack{n \in [0, +\infty] \\ -n \leq m \leq n}} \left(Z_{m,n} - \alpha \right) u_{m,n} \overline{v_{m,n}} + \alpha \int_{\Gamma} \frac{u(\mathbf{x}) \overline{v(\mathbf{x})}}{\sqrt{\frac{a^2}{f^2} - \frac{z^2}{a^2}}} ds_{\mathbf{x}} \end{aligned}$$

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This technique is an adaptation of

- M. Lenoir, A. Tounsi, *The localized finite element method and its application to the two-dimensional sea-keeping problem*, SIAM J. Numer. Anal., 25, pp. 729-752, 1988.

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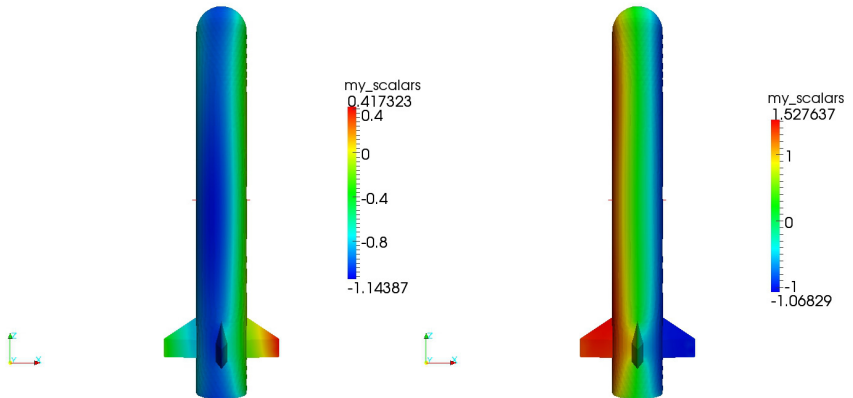
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Theorem

Let u and u_N be the solution of the exact and the approximate problems (N is the order of truncation)

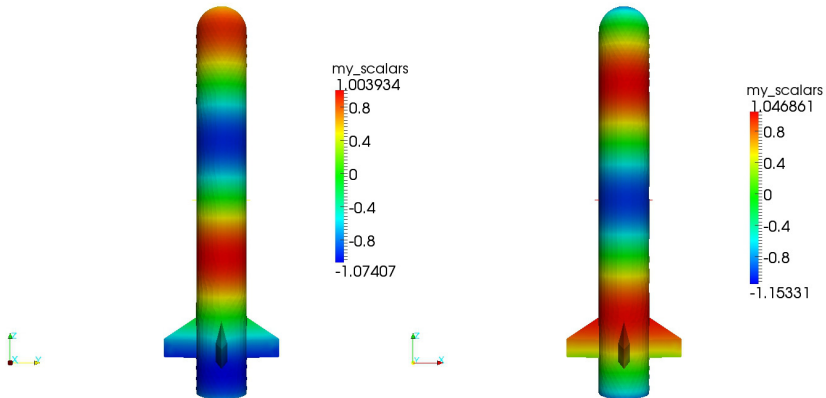
$$\lim_{N \rightarrow +\infty} \|u - u_N\|_{H^1} = 0$$

Real and imaginary part of the scattered phasor for $k = 10$



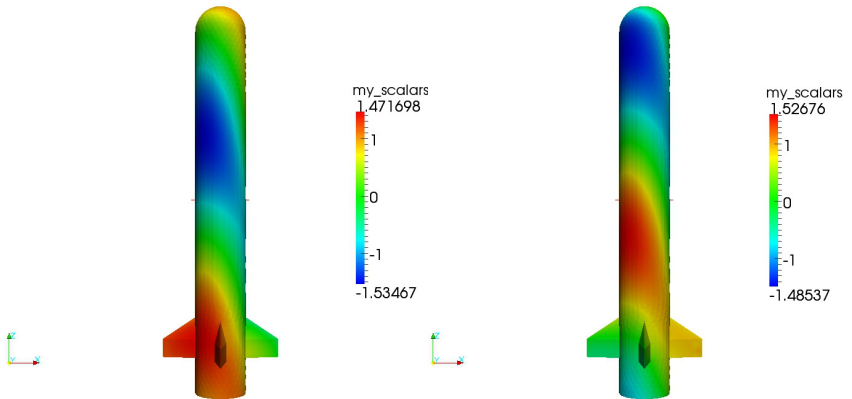
$$\mathbf{k} = (10, 0, 0)$$

Real and imaginary part of the scattered phasor for $k = 10$



$$\mathbf{k} = (0, 0, 10)$$

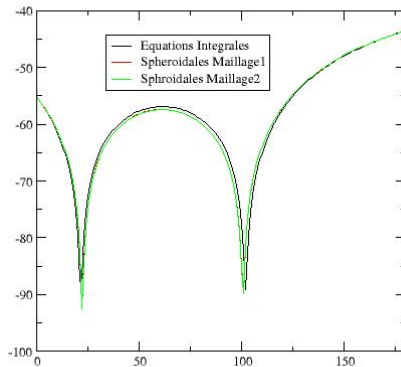
Real and imaginary part of the scattered phasor for $k = 10$



$$\mathbf{k} = (10, 10, 10)/\sqrt{3}$$

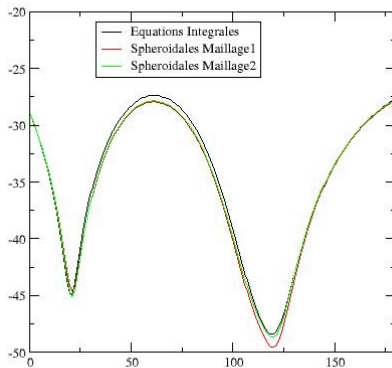
Radar cross-section (RCS) for $\mathbf{k} = (\frac{k}{\sqrt{3}}, \frac{k}{\sqrt{3}}, \frac{k}{\sqrt{3}})$

wave number $k = 1$



Radar cross-section (RCS) for $\mathbf{k} = (\frac{k}{\sqrt{3}}, \frac{k}{\sqrt{3}}, \frac{k}{\sqrt{3}})$

wave number $k = 5$



Radar cross-section (RCS) for $\mathbf{k} = (\frac{k}{\sqrt{3}}, \frac{k}{\sqrt{3}}, \frac{k}{\sqrt{3}})$

wave number $k = 10$

